

Daffodil International University
DEPARTMENT OF INFORMATION & COMMUNICATION ENGINEERING
Dhaka, BANGLADESH.
COURSE NO.: ICE 3242
COURSE TITLE: DIGITAL SIGNAL PROCESSING Lab

Experiment No. 01
Name of Experiment: Generation of Signals

Objectives:

The objectives of this experiment are:

To generate the following discrete-time signals using MATLAB 7.0:

1. Unit impulse signal
2. Unit step signal
3. Unit ramp signal
4. Exponential growing signal
5. Exponential decaying signal
6. Sine signal
7. Cosine signal

To plot the generated signals and observe their amplitude and time characteristics.

THEORY:

A signal is a function that conveys information about a physical phenomenon. A discrete-time signal $x(n)$ is defined only at integer values of the independent variable n . The elementary discrete-time signals used as building blocks in digital signal processing are described below.

1. Unit Impulse Signal

The unit impulse (unit sample) sequence is equal to 1 at $n = 0$ and 0 elsewhere.

$$\delta(n) = \begin{cases} 1, & n = 0 \\ 0, & n \neq 0 \end{cases}$$

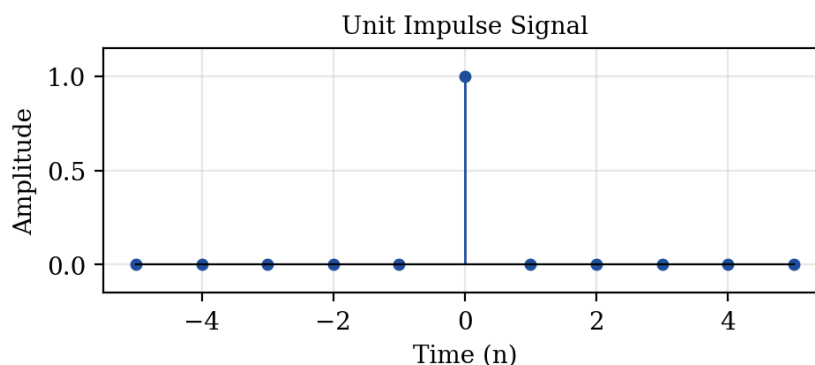


Fig. 1.1 Unit impulse signal $\delta(n)$

2. Unit Step Signal

The unit step sequence is 0 for negative n and 1 for $n \geq 0$. It is the running sum of the unit impulse.

$$u(n) = \begin{cases} 1, & n \geq 0 \\ 0, & n < 0 \end{cases}$$

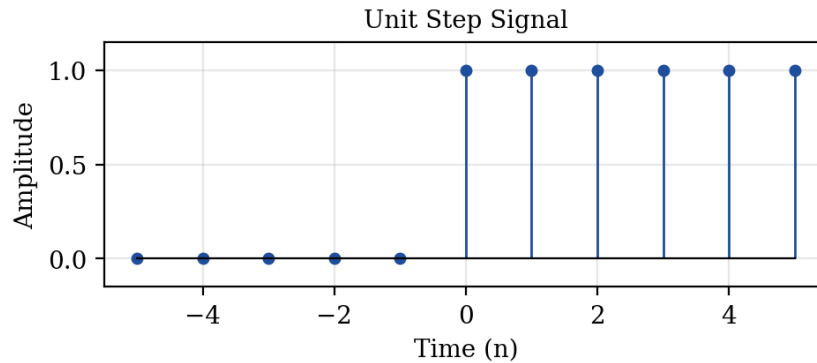


Fig. 1.2 Unit step signal $u(n)$

3. Unit Ramp Signal

The ramp sequence increases linearly with n for $n \geq 0$, where a is the slope (amplitude). It is the running sum of the unit step.

$$r(n) = \begin{cases} a n, & n \geq 0 \\ 0, & n < 0 \end{cases}$$

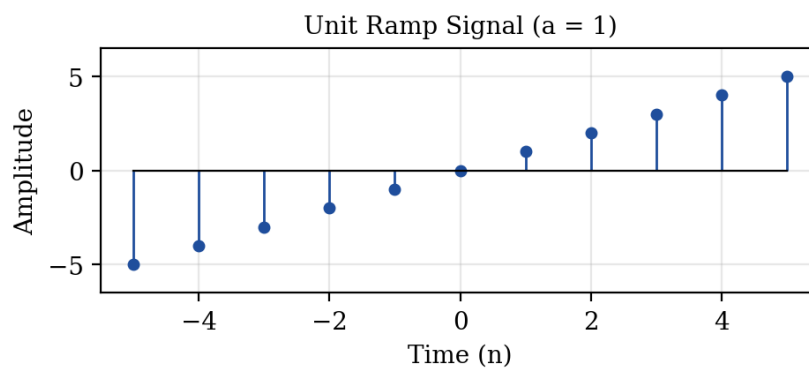


Fig. 1.3 Unit ramp signal $r(n)$

4. Exponential Decaying Signal

A real exponential sequence has the form a^n . When $0 < a < 1$ the amplitude decreases with n , giving an exponential decaying signal.

$$x(n) = a^n, \quad 0 < a < 1$$

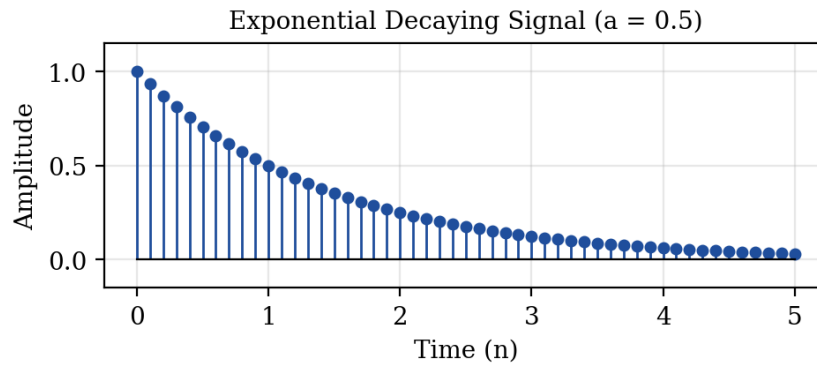


Fig. 1.4 Exponential decaying signal, $a = 0.5$

5. Exponential Growing Signal

When $a > 1$ the real exponential sequence a^n increases without bound as n increases, giving an exponential growing signal.

$$x(n) = a^n, \quad a > 1$$

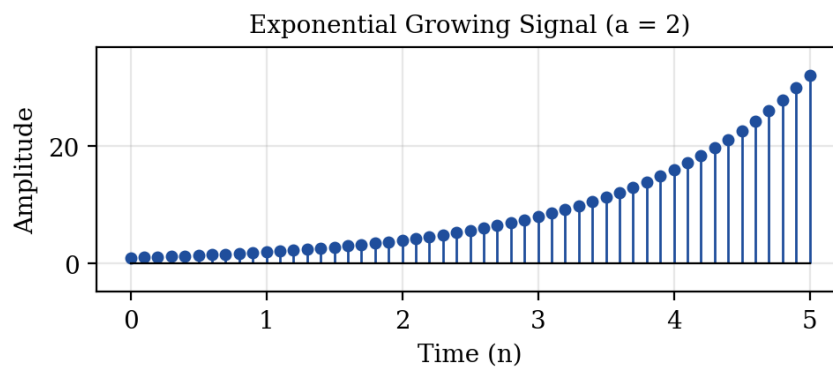


Fig. 1.5 Exponential growing signal, $a = 2$

6. Cosine Signal

A cosine sequence is obtained by sampling a continuous cosine wave. Here A is the amplitude, ω_0 the angular frequency in rad/sample and ϕ the phase. The sequence is periodic only when $\omega_0/2\pi$ is a rational number.

$$x(n) = A \cos(\omega_0 n + \phi) = A \cos(2\pi f_0 n + \phi)$$

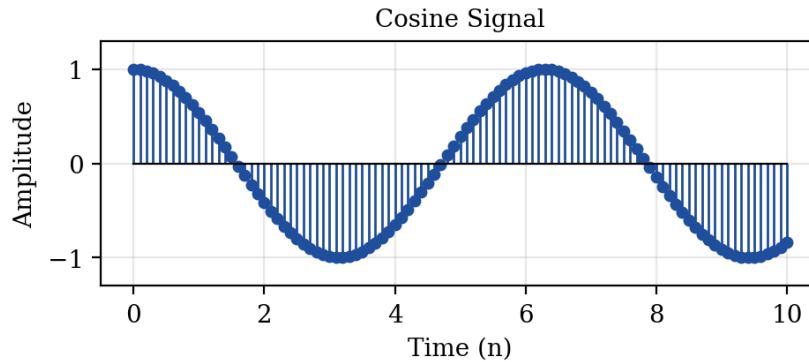


Fig. 1.6 Cosine signal

7. Sine Signal

The sine sequence has the same form as the cosine but is shifted by $\pi/2$ radians: $\sin(\omega_0 n) = \cos(\omega_0 n - \pi/2)$.

$$x(n) = A \sin(\omega_0 n + \phi) = A \sin(2\pi f_0 n + \phi)$$

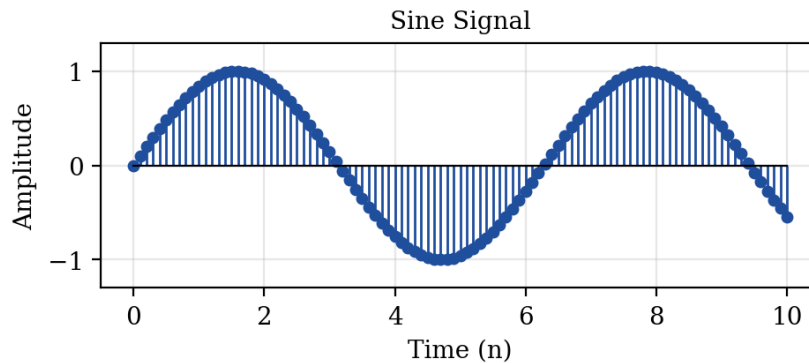


Fig. 1.7 Sine signal

Relationship between the elementary signals

The impulse, step and ramp signals are related as follows:

$$\delta(n) = u(n) - u(n-1), \quad u(n) = \sum_{k=-\infty}^n \delta(k), \quad r(n) = \sum_{k=-\infty}^n u(k-1)$$

MATLAB commands used

- **zeros(1,N)** – creates a row vector of N zeros.
- **ones(1,N)** – creates a row vector of N ones.
- **stem(n,x)** – plots the discrete sequence x against the index n.
- **input(...)** – reads a value entered by the user.

- **xlabel, ylabel, title** – label the axes and the plot.

PROCEDURE:

Apparatus Required

- System with MATLAB 7.0.

Algorithm

1. Get the number of samples.
2. Generate the unit impulse and unit step signals using the ‘ones’ and ‘zeros’ matrix commands.
3. Generate the ramp, sine, cosine and exponential signals using the corresponding general formulas.
4. Plot the graph.

Program

1. Unit Impulse Signal

```
% 1. UNIT IMPULSE SIGNAL
clc;
clear all;
close all;
N=input('Number of Samples = ');
n=-N:1:N
x=[zeros(1,N) 1 zeros(1,N)]
stem(n,x);
xlabel('Time');
ylabel('Amplitude');
title('Impulse Response');
```

2. Unit Step Signal

```
% 2. UNIT STEP SIGNAL
clc;
clear all;
close all;
N=input('Number of Samples = ');
n=-N:1:N
x=[zeros(1,N) 1 ones(1,N)]
stem(n,x);
xlabel('Time');
ylabel('Amplitude');
title('Unit Step Response');
```

3. Unit Ramp Signal

```
% 3. UNIT RAMP SIGNAL
clc;
clear all;
close all;
disp('UNIT RAMP SIGNAL');
N=input('Number of Samples = ');
a=input('Amplitude = ')
n=-N:1:N
x=a*n
stem(n,x);
xlabel('Time');
ylabel('Amplitude');
title('Unit Ramp Response');
```

4. Exponential Decaying Signal

```
% 4. EXPONENTIAL DECAYING SIGNAL
```

```

clc;
clear all;
close all;
disp('EXPONENTIAL DECAYING SIGNAL');
N=input('Number of Samples = ');
a=0.5
n=0:.1:N
x=a.^n
stem(n,x);
xlabel('Time');
ylabel('Amplitude');
title('Exponential Decaying Signal Response');

```

5. Exponential Growing Signal

```

% 5. EXPONENTIAL GROWING SIGNAL
clc;
clear all;
close all;
disp('EXPONENTIAL GROWING SIGNAL');
N=input('Number of Samples = ');
a=2 % a > 1 for a growing exponential
n=0:.1:N
x=a.^n
stem(n,x);
xlabel('Time');
ylabel('Amplitude');
title('Exponential Growing Signal Response');

```

6. Cosine Signal

```

% 6. COSINE SIGNAL
clc;
clear all;
close all;
disp('COSINE SIGNAL');
N=input('Number of Samples = ');
n=0:.1:N
x=cos(n)
stem(n,x);
xlabel('Time');
ylabel('Amplitude');
title('Cosine Signal');

```

7. Sine Signal

```

% 7. SINE SIGNAL
clc;
clear all;
close all;
disp('SINE SIGNAL');
N=input('Number of Samples = ');
n=0:.1:N
x=sin(n)
stem(n,x);
xlabel('Time');
ylabel('Amplitude');
title('Sine Signal');

```

RESULTS:

Run each program in MATLAB, enter the number of samples when prompted, and paste the resulting figure in the space provided below.

S.No.	Signal	Output Waveform (paste figure here)
1	Unit Impulse Signal	
2	Unit Step Signal	
3	Unit Ramp Signal	
4	Exponential Decaying Signal	

S.No.	Signal	Output Waveform (paste figure here)
5	Exponential Growing Signal	
6	Cosine Signal	
7	Sine Signal	

Observations

Conclusion

Thus the unit impulse, unit step, unit ramp, exponential growing, exponential decaying, sine and cosine signals were generated and plotted using MATLAB 7.0.

Exercise:

1. Generate and plot a delayed unit impulse $\delta(n - 3)$ and an advanced unit step $u(n + 2)$ for $-10 \leq n \leq 10$.
2. Modify the exponential programs to use $a = 0.8$ and $a = 1.2$. Plot both on the same figure using subplot and compare their behaviour.

3. Generate a ramp signal with a negative slope ($a = -2$) and explain how the plot differs from $a = 2$.
4. Generate a sine wave of frequency 5 Hz sampled at $f_s = 100$ Hz for 1 second. Plot it using both plot and stem.
5. Generate $x(n) = 2 \sin(0.2\pi n) + \cos(0.5\pi n)$ for $0 \leq n \leq 40$ and plot it. Is the resulting signal periodic? Justify.
6. Express the unit step $u(n)$ in terms of unit impulses and verify it by plotting the running sum of $\delta(n)$ using cumsum.